Code No: 181AJ **R22**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech I Year I Semester Examinations, March/April - 2023

ENGINEERING CHEMISTRY

(Common to EEE, CSE, IT, CSIT, CE(SE), CSE(CS), CSE(DS), CSD)

Time: 3 Hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

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PART- A

(10 Marks)

1.a)	What are the methods used for disinfection of water.	[1]
b)	What is importance of ion exchange method of water treatment?	[1]
c)	Define cathodic protection.	[1]
d)	What are the applications of fuel cells?	[1]
e)	Give applications of terelene.	[1]
f)	Give examples for thermosetting polymers.	[1]
g)	What is the composition of LPG?	[1]
h)	Define HCV.	[1]
i)	Give applications of shape memory materials.	[1]
j)	Define lubricant.	[1]

PART-B

(50 Marks)

2.a)	Define portable water, give its specifications.	
b)	Explain determination of F^- ion by ion- selective electrode method.	[5+5]
	OR	
3.a)	Differentiate internal and external treatment of water.	
b)	Explain reverse osmosis method and its importance.	[5+5]
4.	What are the basic requirements for commercial batteries, explain construction, working principle and applications of Zn-air batteries.	[10]
	OR	
5.a)	What are the causes and effects of corrosion?	
b)	Explain Galvanic and pitting corrosion.	[5+5]
6.a)	Explain steps involved in free radical polymerization mechanism with suitable example.	
b)	Give brief note on Bakelite.	[5+5]
	OR	

7.a) Give brief note on Fibre reinforced plastics and its applications.
b) Explain properties and applications of synthetic rubbers. [5+5]

8. Compare proximate and ultimate analysis of fuel and give its significance. [10]

OR

9.a) Describe Fischer-Tropsch's process.
b) Give brief note on Trans esterification. [5+5]

10. Explain various smart materials and their importance. [10]

OR

11.a) Explain mechanism of lubrication.
b) Explain any three important properties of lubricants. [5+5]

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